

NPVegData Analysis 2025

2025-10-30

```
library(tidyverse)
library(janitor)
library(readr)
library(readxl)
library(gt)
library(purrr)
```

Housekeeping

Load Data

```
#This year's data
alldata_now <- read_csv("2026-ALL.csv")

## Rows: 1576 Columns: 4
## -- Column specification -----
## Delimiter: ","
## chr (3): species, transect, cover_type
## dbl (1): cover
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

#Past data
alldata_25 <- read_csv("2025-ALL.csv")

## Rows: 1570 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (3): Species, Transect, Cover Type
## dbl (2): Cover, Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

alldata_23 <- read_csv("2023-ALL.csv")

## Rows: 999 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (3): Species, Transect, Cover type
## dbl (2): Cover, Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

alldata_22 <- read_csv("2022-ALL.csv")

## Rows: 1185 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (3): Species, Transect, Cover Type
## dbl (2): Cover, Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
alldata_21 <- read_csv("2021-ALL.csv")

```

```

## Rows: 991 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (3): Species, Transect, Cover Type
## dbl (2): Cover, Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

Clean data

```

#read in Name change csv which is used to standardize transect ID names
name_change <- read_csv("NameChange.csv") %>%
  clean_names() # -> startid, endid

## Rows: 18 Columns: 2
## -- Column specification -----
## Delimiter: ","
## chr (2): startid, endid
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
#Standardize cover_type to title case and remove extra spaces. Change transect IDs
alldata_now <- alldata_now %>%
  clean_names() %>%
  mutate(cover_type = str_to_title(str_trim(cover_type))) %>%
  left_join(name_change, by = c("transect" = "startid")) %>%
  mutate(transect = coalesce(endid, transect)) %>%
  select(-endid)

alldata_25 <- alldata_25 %>%
  clean_names() %>%
  mutate(cover_type = str_to_title(str_trim(cover_type))) %>%
  left_join(name_change, by = c("transect" = "startid")) %>%
  mutate(transect = coalesce(endid, transect)) %>%
  select(-endid)

alldata_23 <- alldata_23 %>%
  clean_names() %>%
  mutate(cover_type = str_to_title(str_trim(cover_type))) %>%
  left_join(name_change, by = c("transect" = "startid")) %>%

```

```

mutate(transect = coalesce(endid, transect)) %>%
select(-endid)

alldata_22 <- alldata_22 %>%
  clean_names() %>%
  mutate(cover_type = str_to_title(str_trim(cover_type))) %>%
  left_join(name_change, by = c("transect" = "startid")) %>%
  mutate(transect = coalesce(endid, transect)) %>%
  select(-endid)

alldata_21 <- alldata_21 %>%
  clean_names() %>%
  mutate(cover_type = str_to_title(str_trim(cover_type))) %>%
  left_join(name_change, by = c("transect" = "startid")) %>%
  mutate(transect = coalesce(endid, transect)) %>%
  select(-endid)

```

Current Year - Core Metrics

Percent Cover

```

##Gross Cover

# Calculate cover for each category
native_cover <- alldata_now %>%
  filter(cover_type == "Native") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

exotic_cover <- alldata_now %>%
  filter(cover_type == "Exotic") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

# Calculate cover for specific subcategories based on species column
thatch_cover <- alldata_now %>%
  filter(species == "Thatch") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

mowed_thatch_cover <- alldata_now %>%
  filter(species == "Mowed_thatch") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

bare_ground_cover <- alldata_now %>%
  filter(species == "Bare_ground") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

# Calculate remaining "Other" (cover_type = "Other" but not the three specific species)
remaining_other_cover <- alldata_now %>%

```

```

filter(cover_type == "Other" & !species %in% c("Thatch", "Mowed_thatch", "Bare_ground")) %>%
summarize(total = sum(cover, na.rm = TRUE)) %>%
pull(total)

# Total cover
total_cover <- alldata_now %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

# Calculate percentages
percent_native <- (native_cover / total_cover) * 100
percent_exotic <- (exotic_cover / total_cover) * 100
percent_thatch <- (thatch_cover / total_cover) * 100
percent_mowed_thatch <- (mowed_thatch_cover / total_cover) * 100
percent_bare_ground <- (bare_ground_cover / total_cover) * 100
percent_remaining_other <- (remaining_other_cover / total_cover) * 100

## Relative Cover

# Total vegetative cover (exclude 'other')
total_veg_cover <- alldata_now %>%
  filter(cover_type != "Other") %>% # only actual vegetation
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

# Relative percentages
percent_relative_native <- (native_cover / total_veg_cover) * 100
percent_relative_exotic <- (exotic_cover / total_veg_cover) * 100

## Results

# Create a tibble with all categories
total_cover_pct <- tibble(
  Cover_Type = c("Native", "Exotic", "Other", "Other", "Other", "Other"),
  Subcategory = c("Native", "Exotic", "Thatch", "Mowed Thatch", "Bare Ground", "Other"),
  Percent_Cover = c(percent_native, percent_exotic, percent_thatch,
                    percent_mowed_thatch, percent_bare_ground, percent_remaining_other)
)

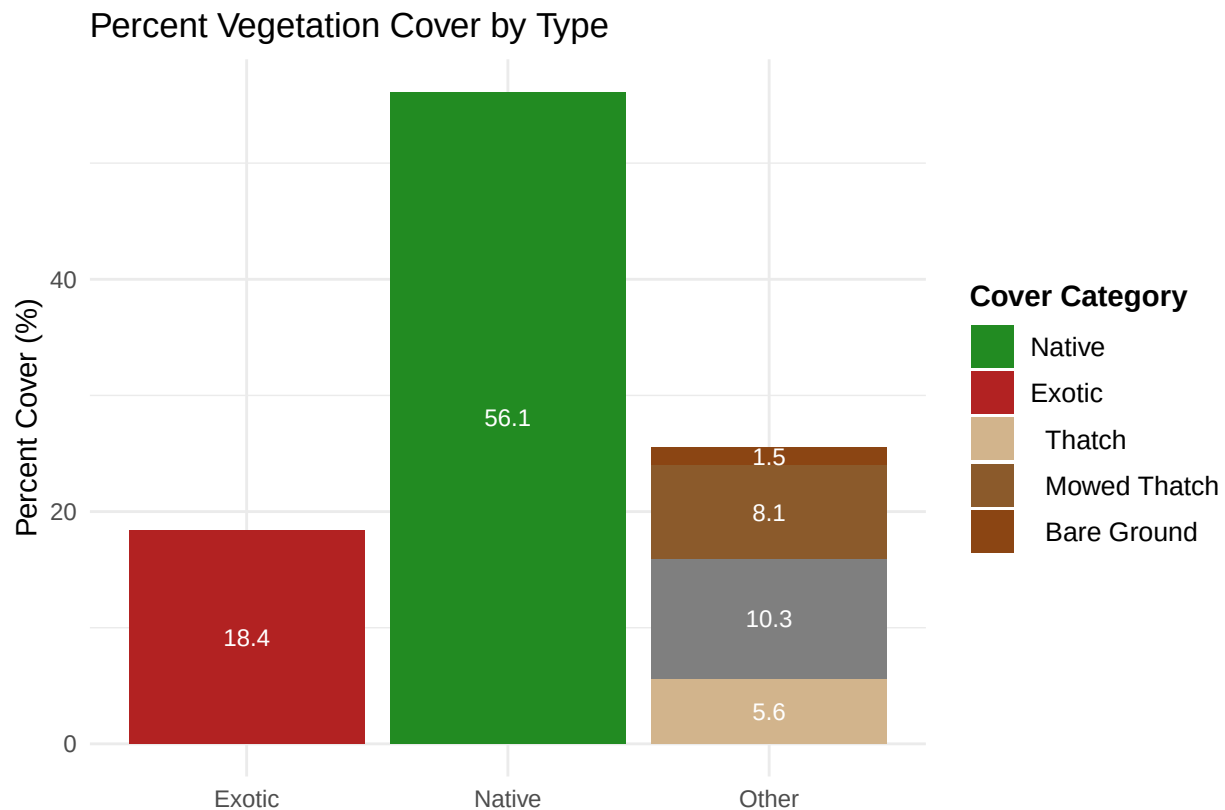
# Plot it
ggplot(total_cover_pct, aes(x = Cover_Type, y = Percent_Cover, fill = Subcategory)) +
  geom_col() +
  labs(
    title = "Percent Vegetation Cover by Type",
    y = "Percent Cover (%)",
    x = "",
    fill = "Cover Category"
  ) +
  theme_minimal() +
  geom_text(aes(label = round(Percent_Cover, 1)),
            position = position_stack(vjust = 0.5),

```

```

      size = 3.2,
      color = "white") +
  theme(
    legend.position = "right",
    legend.text = element_text(size = 10),
    legend.title = element_text(size = 11, face = "bold")
  ) +
  scale_fill_manual(
    values = c("Native" = "forestgreen",
              "Exotic" = "firebrick",
              "Thatch" = "tan",
              "Mowed Thatch" = "tan4",
              "Bare Ground" = "saddlebrown",
              "other" = "gray"),
    breaks = c("Native", "Exotic", "Thatch", "Mowed Thatch", "Bare Ground", "other"),
    labels = c("Native", "Exotic", " Thatch", " Mowed Thatch", " Bare Ground", " other")
  ) +
  guides(fill = guide_legend(
    title = "Cover Category",
    override.aes = list(size = 5)
  ))

```



```

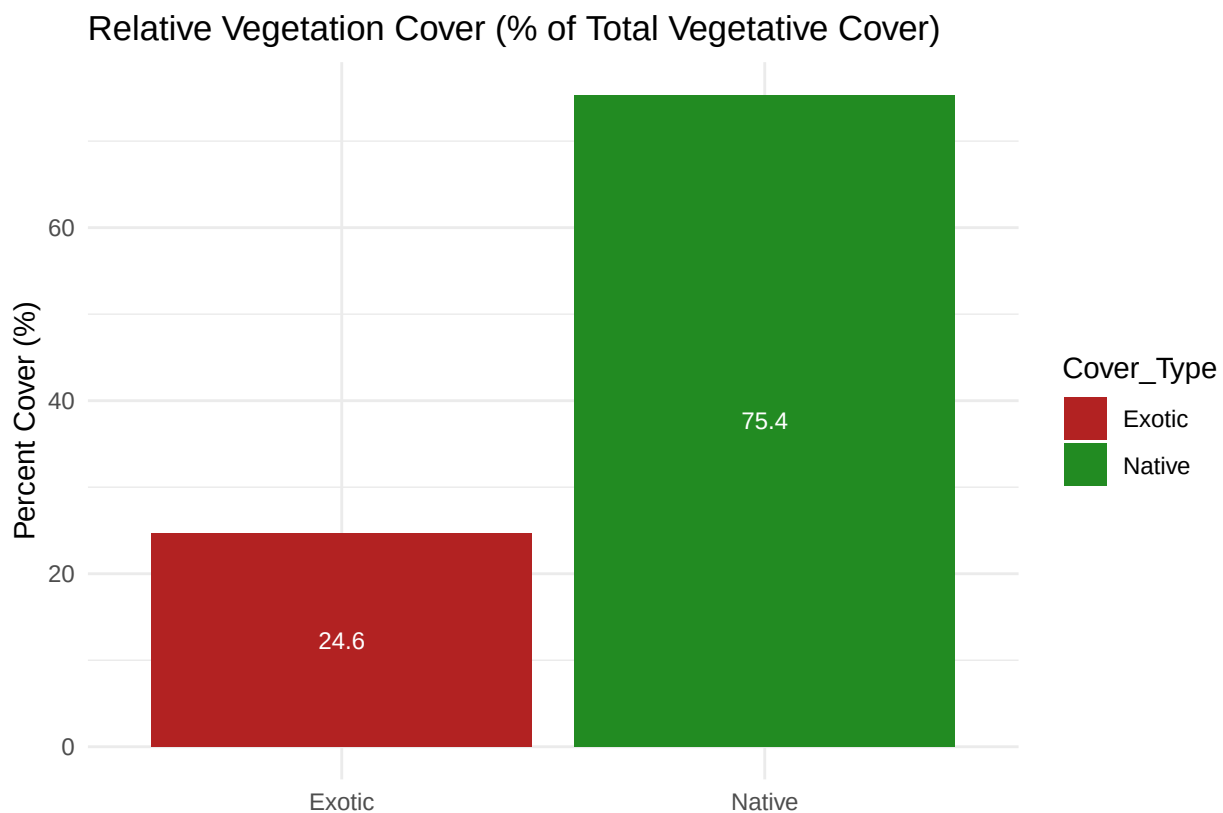
#Create tibble for relative percent cover

relative_cover_pct <- tibble(
  Cover_Type = c("Native", "Exotic"),
  Percent_Cover = c(percent_relative_native, percent_relative_exotic)
)

```

```
#Plot relative percent cover

ggplot(relative_cover_pct, aes(x = Cover_Type, y = Percent_Cover, fill = Cover_Type)) +
  geom_col() +
  geom_text(
    aes(
      y = Percent_Cover / 2,          # place label halfway up the bar
      label = round(Percent_Cover, 1)
    ),
    color = "white",
    size = 3.2
  ) +
  labs(
    title = "Relative Vegetation Cover (% of Total Vegetative Cover)",
    y = "Percent Cover (%)",
    x = ""
  ) +
  theme_minimal() +
  scale_fill_manual(values = c("Native" = "forestgreen", "Exotic" = "firebrick"))
```



Percent Cover By Transect

```
# Get all unique transects

transects <- unique(alldata_now$transect)

# Initialize empty tibble for results
```

```

all_transect_summary <- tibble()

# Loop through each transect

for (VP in transects) {

  # Filter data for this transect

  VP_data <- alldata_now %>%
  filter(transect == VP)

  # Total cover

  total_cover_VP <- sum(VP_data$cover, na.rm = TRUE)

  # Native cover

  native_cover_VP <- VP_data %>%
  filter(cover_type == "Native") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

  # Exotic cover

  exotic_cover_VP <- VP_data %>%
  filter(cover_type == "Exotic") %>%
  summarize(total = sum(cover, na.rm = TRUE)) %>%
  pull(total)

  # other cover

  other_cover_VP <- total_cover_VP - (native_cover_VP + exotic_cover_VP)

  # Percent covers

  percent_native_VP <- (native_cover_VP / total_cover_VP) * 100
  percent_exotic_VP <- (exotic_cover_VP / total_cover_VP) * 100
  percent_other_VP <- (other_cover_VP / total_cover_VP) * 100

  # Relative cover (exclude Other)

  total_veg_cover_VP <- native_cover_VP + exotic_cover_VP
  percent_relative_native_VP <- (native_cover_VP / total_veg_cover_VP) * 100
  percent_relative_exotic_VP <- (exotic_cover_VP / total_veg_cover_VP) * 100

  # Store results

  VP_summary <- tibble(
  Transect = VP,
  Percent_Native = percent_native_VP,
  Percent_Exotic = percent_exotic_VP,
  Percent_Other = percent_other_VP,
  Relative_Native = percent_relative_native_VP,

```

```

Relative_Exotic = percent_relative_exotic_VP
)

# Append

all_transect_summary <- bind_rows(all_transect_summary, VP_summary)
}

##Graph it

# Reshape for plotting

all_transect_long <- all_transect_summary %>%
pivot_longer(
cols = c(Percent_Native, Percent_Exotic, Percent_Other),
names_to = "Cover_Type",
values_to = "Percent_Cover"
) %>%
mutate(Cover_Type = case_when(
Cover_Type == "Percent_Native" ~ "Native",
Cover_Type == "Percent_Exotic" ~ "Exotic",
Cover_Type == "Percent_Other" ~ "Other"
))

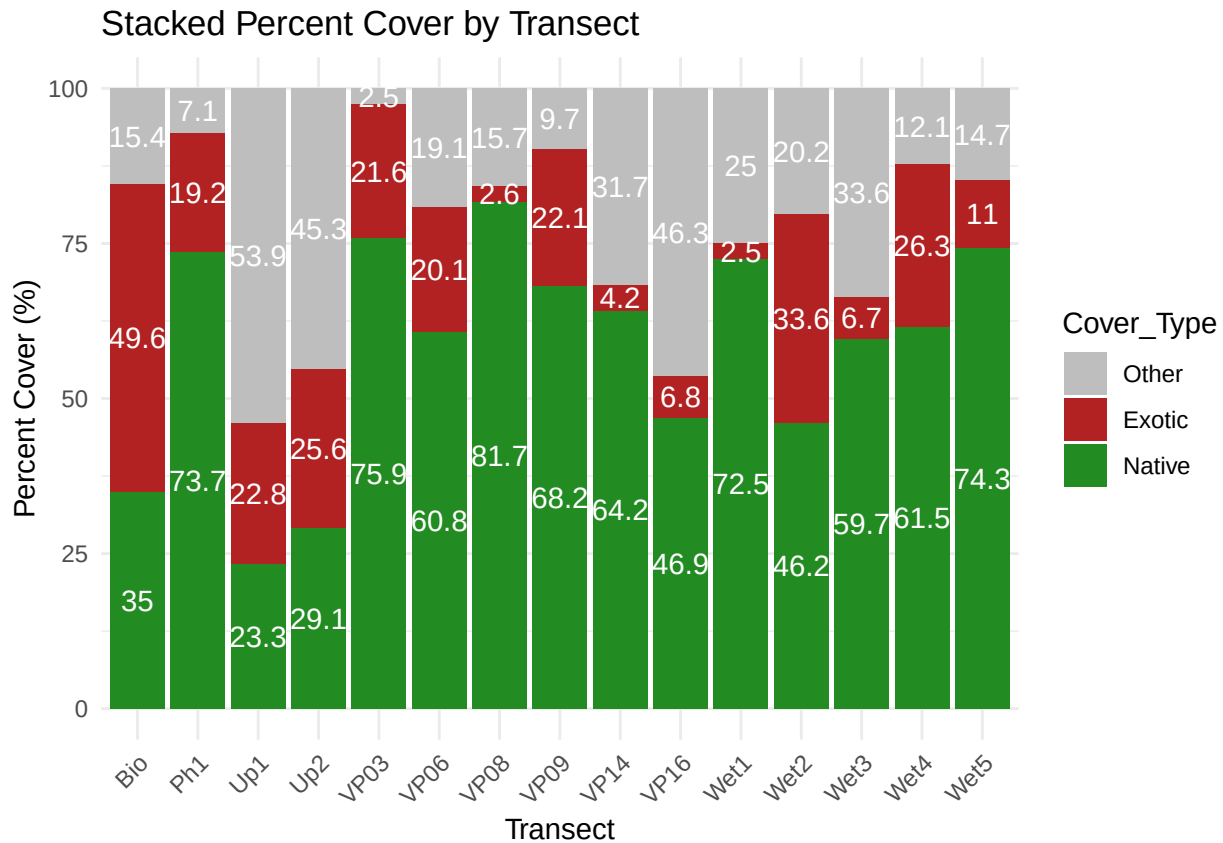
# Set stacking order

all_transect_long$Cover_Type <- factor(all_transect_long$Cover_Type, levels = c("Other", "Exotic", "Native"))

# Plot stacked bars

ggplot(all_transect_long, aes(x = Transect, y = Percent_Cover, fill = Cover_Type)) +
geom_col() +
labs(
title = "Stacked Percent Cover by Transect",
y = "Percent Cover (%)",
x = "Transect"
) +
scale_fill_manual(values = c("Native" = "forestgreen", "Exotic" = "firebrick", "Other" = "gray")) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))+
  geom_text(aes(label = round(Percent_Cover, 1)), # round to 1 decimal
            position = position_stack(vjust = 0.5), # place labels in the middle of each bar
            color = "white", size = 4)

```



```
# Relative cover (Native and Exotic only)

relative_cover_long <- all_transect_summary %>%
select(Transect, Relative_Native, Relative_Exotic) %>%
pivot_longer(
  cols = c(Relative_Native, Relative_Exotic),
  names_to = "Cover_Type",
  values_to = "Percent_Cover"
) %>%
mutate(Cover_Type = case_when(
  Cover_Type == "Relative_Native" ~ "Native",
  Cover_Type == "Relative_Exotic" ~ "Exotic"
))

# Set order

relative_cover_long$Cover_Type <- factor(relative_cover_long$Cover_Type, levels = c("Exotic", "Native"))

# Plot

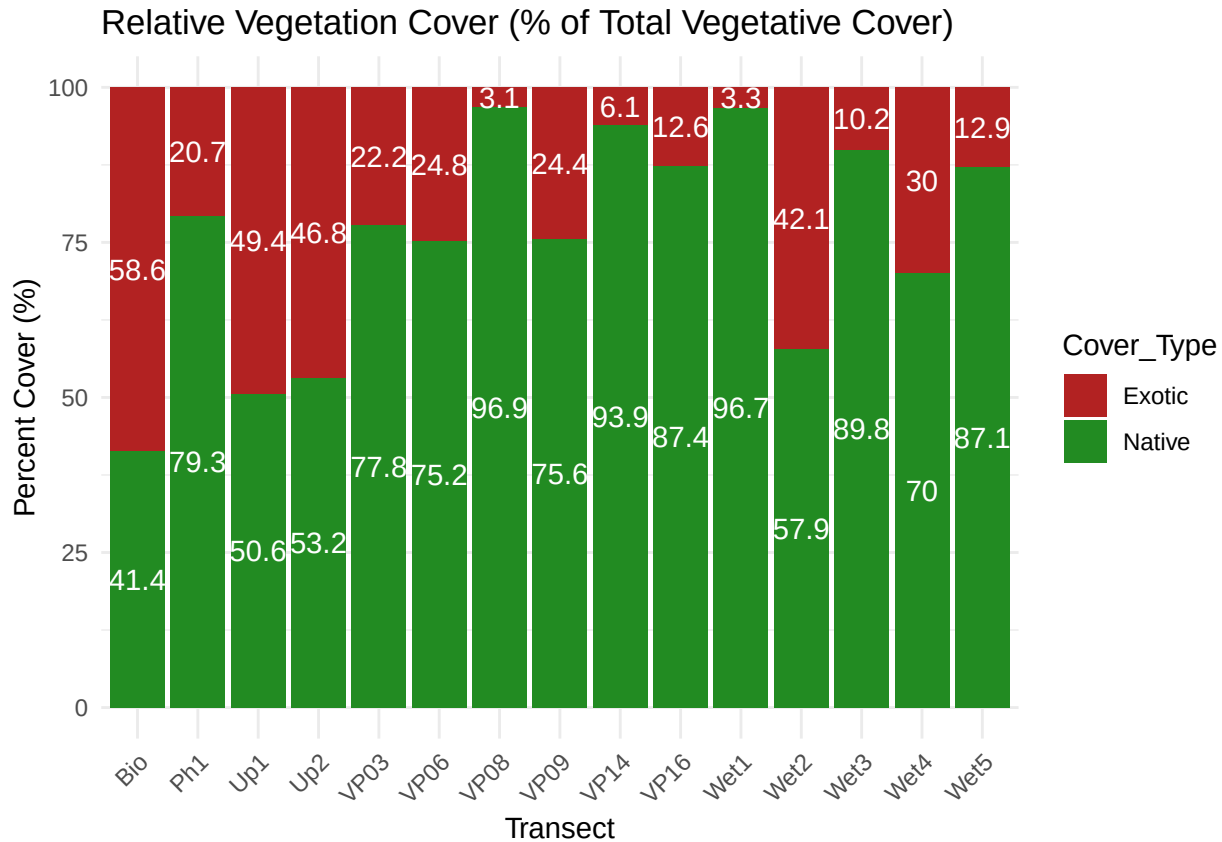
ggplot(relative_cover_long, aes(x = Transect, y = Percent_Cover, fill = Cover_Type)) +
  geom_col() +
  geom_text(aes(label = round(Percent_Cover, 1)), # round to 1 decimal
            position = position_stack(vjust = 0.5), # place labels in the middle of each bar
            color = "white", size = 4) +
  labs(
    title = "Relative Vegetation Cover (% of Total Vegetative Cover)",

```

```

y = "Percent Cover (%)",
x = "Transect"
) +
scale_fill_manual(values = c("Native" = "forestgreen", "Exotic" = "firebrick")) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



Problem weeds

```

# Top 10 most abundant exotic species by total cover and percent of exotic cover

total_exotic_cover <- sum(alldata_now$cover[alldata_now$cover_type == "Exotic"], na.rm = TRUE)

top_exotic_species <- alldata_now %>%
  filter(cover_type == "Exotic") %>%
  group_by(species) %>%
  summarize(Total_Cover = sum(cover, na.rm = TRUE)) %>%
  mutate(Percent_of_Exotic = (Total_Cover / total_exotic_cover) * 100) %>%
  arrange(desc(Total_Cover)) %>%
  slice_head(n = 10)

## Listed
top_exotic_species

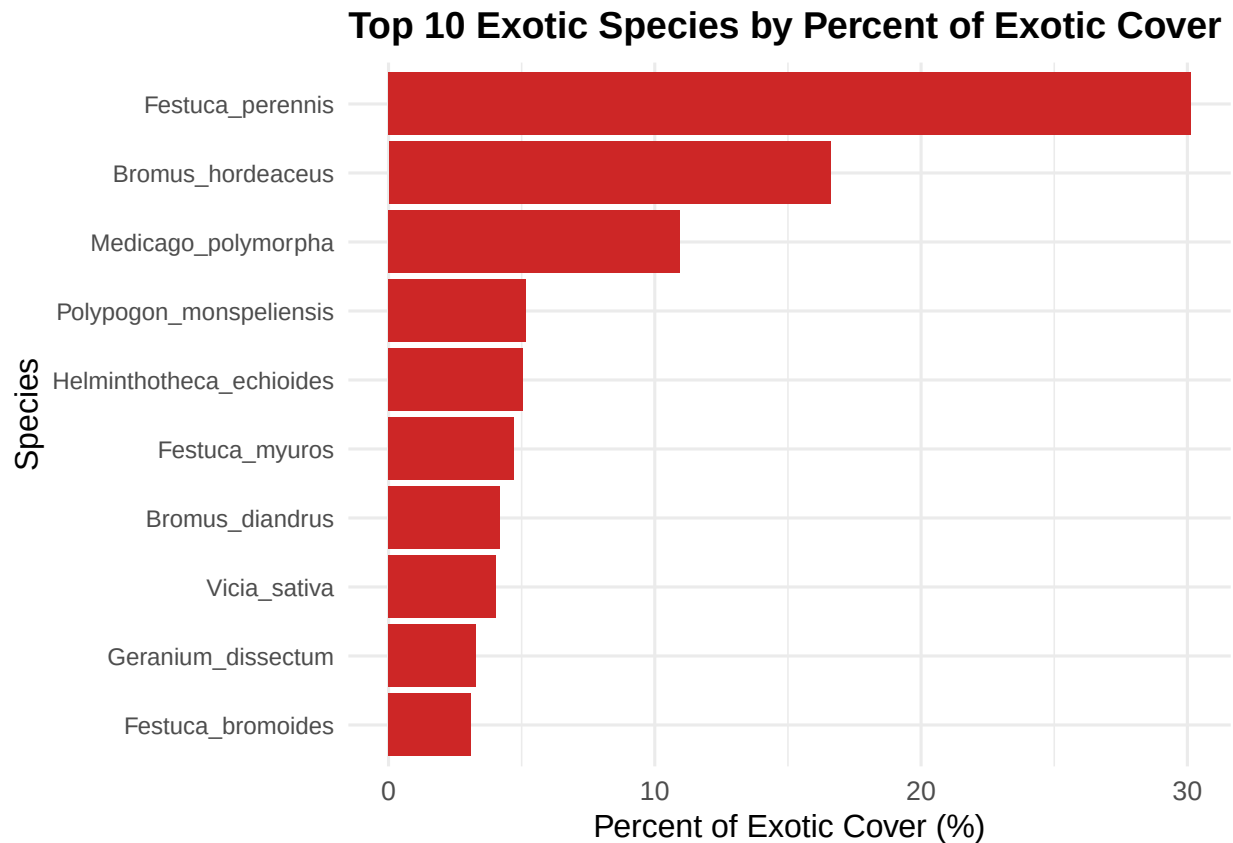
## # A tibble: 10 x 3
##   species                Total_Cover Percent_of_Exotic

```

##	<chr>	<dbl>	<dbl>
##	1 Festuca_perennis	1519	30.1
##	2 Bromus_hordeaceus	837	16.6
##	3 Medicago_polymorpha	552	10.9
##	4 Polypogon_monspeliensis	261	5.18
##	5 Helminthotheca_echioides	254	5.04
##	6 Festuca_myuros	237	4.70
##	7 Bromus_diandrus	212	4.20
##	8 Vicia_sativa	204	4.05
##	9 Geranium_dissectum	165	3.27
##	10 Festuca_bromoides	157	3.11

##Graphed

```
ggplot(top_exotic_species,
  aes(x = reorder(species, Percent_of_Exotic),
    y = Percent_of_Exotic)) +
  geom_col(fill = "firebrick3") +
  coord_flip() +
  labs(
    title = "Top 10 Exotic Species by Percent of Exotic Cover",
    x = "Species",
    y = "Percent of Exotic Cover (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.y = element_text(size = 9),
    plot.title = element_text(face = "bold")
  )
```



Past Data - Core metrics

Automation Housekeeping

```
# ---- MASTER DATA LIST[ADD NEW YEAR HERE] ----
all_data_list <- list(
  `2021` = alldata_21,
  `2022` = alldata_22,
  `2023` = alldata_23,
  `2025` = alldata_25,
  `2026` = alldata_now      # replace with alldata_27 next year etc.
)

# ----- AUTOMATED CALCULATIONS AND GRAPHING BELOW -----

##Pre-automation step
years_used      <- sort(as.integer(names(all_data_list)))
year_range_label <- paste0(min(years_used), "-", max(years_used))
n_top <- 5      # for the exotic species plot

##Pre-automation styling
cover_colors <- c(Native = "forestgreen", Exotic = "firebrick", Other = "gray70")
```

Percent Cover

```
calc_cover_stats <- function(df) {

  total_cover <- sum(df$cover, na.rm = TRUE)
  native_cover <- sum(df$cover[df$cover_type == "Native"], na.rm = TRUE)
  exotic_cover <- sum(df$cover[df$cover_type == "Exotic"], na.rm = TRUE)
  other_cover <- total_cover - (native_cover + exotic_cover)

  # Percentages of total
  percent_native <- 100 * native_cover / total_cover
  percent_exotic <- 100 * exotic_cover / total_cover
  percent_other <- 100 * other_cover / total_cover

  # Relative percentages (no "other")
  veg_total <- native_cover + exotic_cover
  percent_relative_native <- 100 * native_cover / veg_total
  percent_relative_exotic <- 100 * exotic_cover / veg_total

  tibble(
    Percent_Native = percent_native,
    Percent_Exotic = percent_exotic,
    Percent_Other = percent_other,
    Relative_Native = percent_relative_native,
    Relative_Exotic = percent_relative_exotic
  )
}

# Automatically run for all years
all_years_summary <- imap_dfr(
  all_data_list,
  ~ calc_cover_stats(.x) %>% mutate(year = as.integer(.y))
)

# Create singular combined data set
alldata_all <- bind_rows(
  imap(all_data_list, ~ mutate(.x, year = as.integer(.y)))
)

#Results

# Results Table
all_years_summary

## # A tibble: 5 x 6
##   Percent_Native Percent_Exotic Percent_Other Relative_Native Relative_Exotic
##   <dbl>          <dbl>          <dbl>          <dbl>          <dbl>
## 1         40.2         14.7         45.1         73.2         26.8
## 2         49.9         15.8         34.2         75.9         24.1
## 3         49.3         24.9         25.8         66.5         33.5
## 4         39.5         13.4         47.0         74.6         25.4
## 5         56.1         18.4         25.5         75.4         24.6
## # i 1 more variable: year <int>
```

```

#Results Graphs

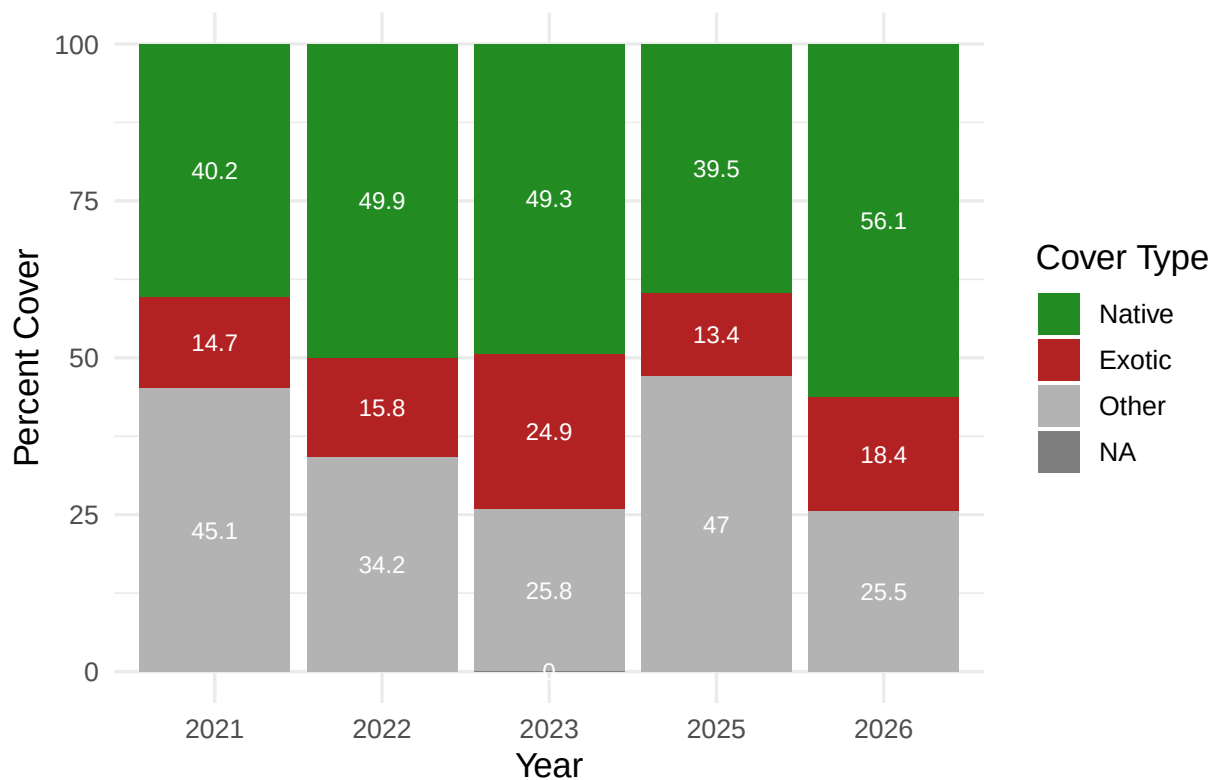
# Summarize total cover per cover type per year
cover_summary <- alldata_all %>%
  group_by(year, cover_type) %>%
  summarize(total_cover = sum(cover, na.rm = TRUE), .groups = "drop") %>%
  group_by(year) %>%
  mutate(percent_cover = (total_cover / sum(total_cover)) * 100)

# Order cover types for consistent stacking
cover_summary$cover_type <- factor(
  cover_summary$cover_type,
  levels = c("Native", "Exotic", "Other")
)

# Plot 1: Total Percent Cover-----
ggplot(cover_summary, aes(x = factor(year), y = percent_cover, fill = cover_type)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(percent_cover, 1)),
    position = position_stack(vjust = 0.5),
    size = 3.2,
    color = "white") +
  labs(
    title = "Total Percent Cover by Year (2021-2025)",
    x = "Year",
    y = "Percent Cover",
    fill = "Cover Type"
  ) +
  theme_minimal(base_size = 13) +
  scale_fill_manual(values = c("forestgreen", "firebrick", "gray70"))

```

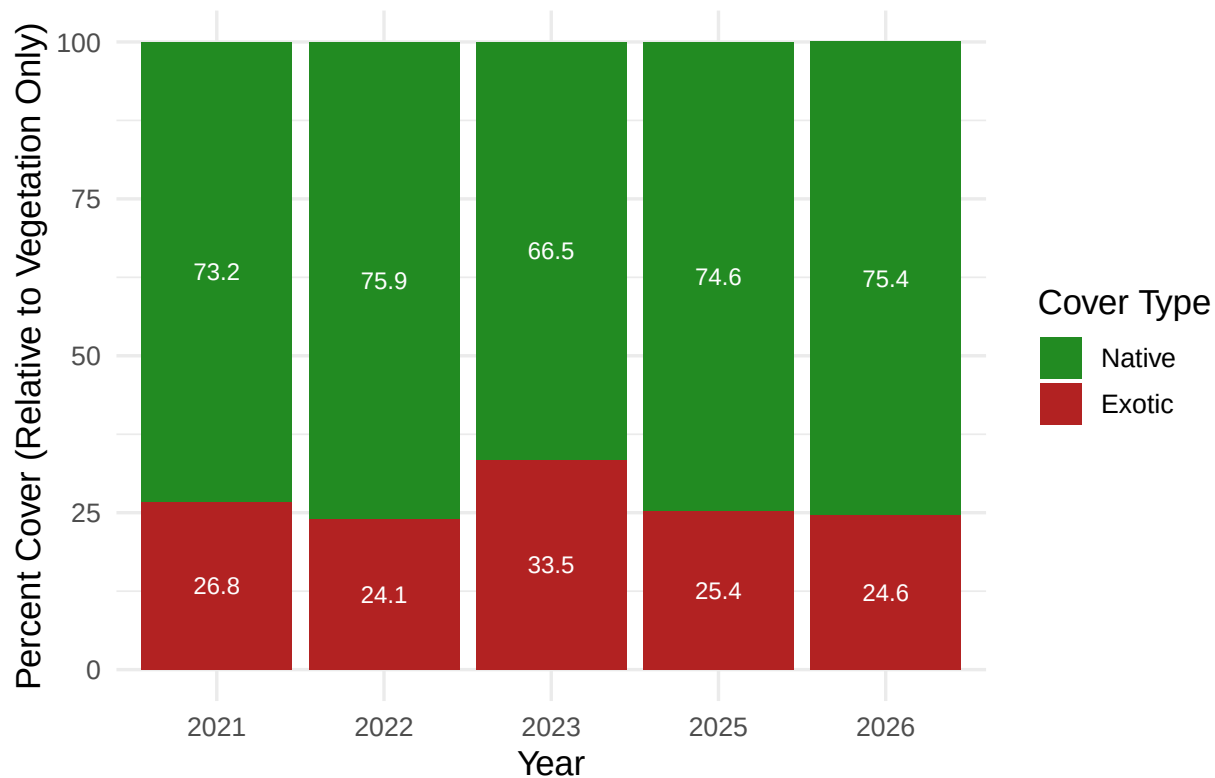
Total Percent Cover by Year (2021–2025)



```
# Plot 2: Relative Percent Cover (Native vs Exotic only) ----
relative_summary <- cover_summary %>%
  filter(cover_type %in% c("Native", "Exotic")) %>%
  group_by(year) %>%
  mutate(percent_relative = (total_cover / sum(total_cover)) * 100)

ggplot(relative_summary, aes(x = factor(year), y = percent_relative, fill = cover_type)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(percent_relative, 1)),
            position = position_stack(vjust = 0.5),
            size = 3.2,
            color = "white") +
  labs(
    title = "Percent of Vegetation Cover by Year (2021-2025, Native vs Exotic)",
    x = "Year",
    y = "Percent Cover (Relative to Vegetation Only)",
    fill = "Cover Type"
  ) +
  theme_minimal(base_size = 13) +
  scale_fill_manual(values = cover_colors)
```

Percent of Vegetation Cover by Year (2021–2025, Native vs



Species Richness

```
calc_species_count <- function(df) {
  df %>%
    summarize(
      n_total_species = n_distinct(species[!is.na(species)]),
      n_native_species = n_distinct(species[cover_type == "Native" & !is.na(species)]),
      n_exotic_species = n_distinct(species[cover_type == "Exotic" & !is.na(species)])
    )
}

species_counts <- imap_dfr(
  all_data_list,
  ~ calc_species_count(.x) %>% mutate(year = as.integer(.y))
)

# ---- Plot: Stacked Species Counts (Native + Exotic Only) ----

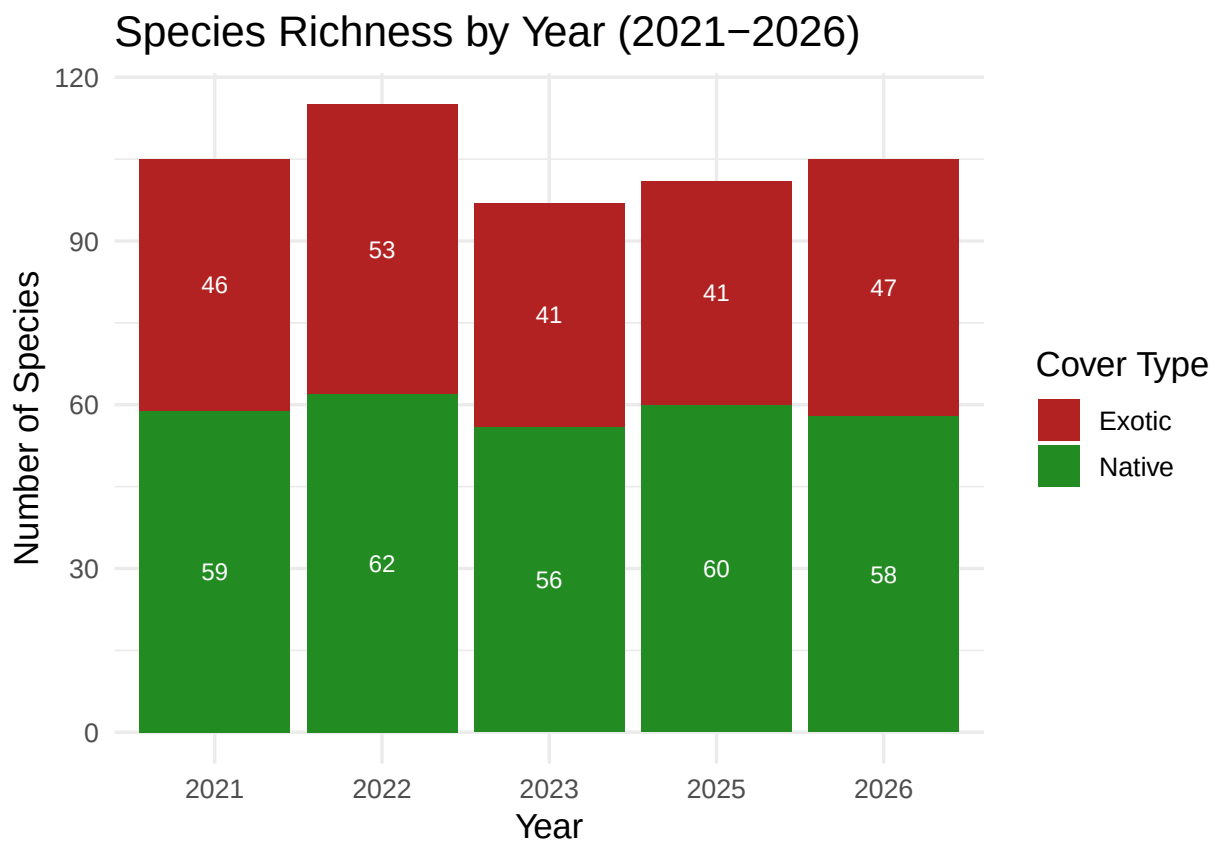
species_counts_long <- species_counts %>%
  pivot_longer(
    cols = c(n_native_species, n_exotic_species),
    names_to = "cover_type",
    values_to = "n_species"
  ) %>%
  mutate(cover_type = recode(cover_type,
```

```

    n_native_species = "Native",
    n_exotic_species = "Exotic"))

ggplot(species_counts_long, aes(x = factor(year), y = n_species, fill = cover_type)) +
  geom_col() +
  geom_text(aes(label = n_species),
            position = position_stack(vjust = 0.5),
            size = 3.2,
            color = "white") +
  scale_fill_manual(values = cover_colors) +
  labs(
    title = paste0("Species Richness by Year (", year_range_label, ")"),
    x = "Year",
    y = "Number of Species",
    fill = "Cover Type"
  ) +
  theme_minimal(base_size = 13)

```



Problem Species over time

```

calc_top5_exotics <- function(df) {
  df_ex <- df %>% filter(cover_type == "Exotic")
  total_ex <- sum(df_ex$cover, na.rm = TRUE)

  df_ex %>%
    group_by(species) %>%

```

```

    summarize(total_cover = sum(cover, na.rm = TRUE), .groups = "drop") %>%
    arrange(desc(total_cover)) %>%
    slice_head(n = 5) %>%
    mutate(percent_of_exotic = 100 * total_cover / total_ex)
}

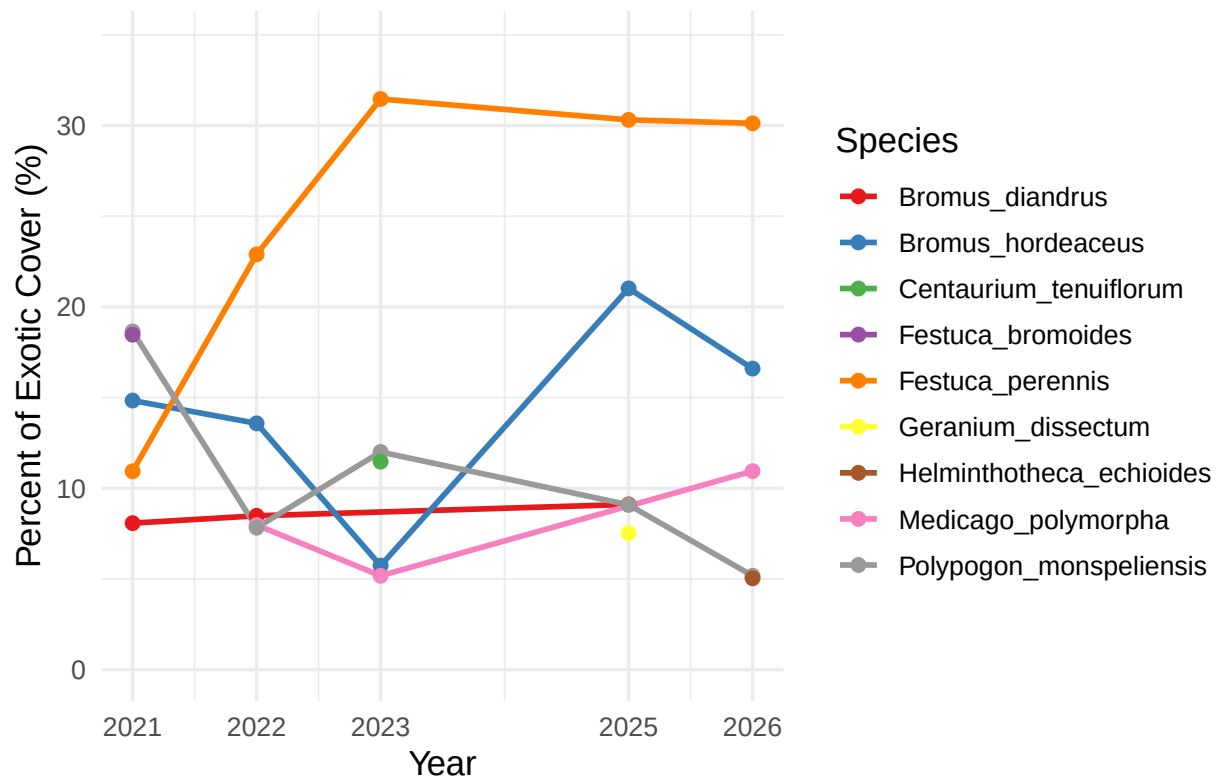
top5_trends <- imap_dfr(all_data_list, ~ {
  calc_top5_exotics(.x) %>% mutate(year = as.integer(.y))
})

# Plot-----
ggplot(top5_trends, aes(x = year, y = percent_of_exotic, color = species, group = species)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(
    title = "Top 5 Exotic Species Each Year: Percent of Total Exotic Cover",
    x = "Year",
    y = "Percent of Exotic Cover (%)",
    color = "Species"
  ) +
  theme_minimal(base_size = 13) +
  scale_x_continuous(breaks = years_used) +
  scale_color_brewer(palette = "Set1") +
  ylim(0, max(top5_trends$percent_of_exotic, na.rm = TRUE) * 1.1)

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Top 5 Exotic Species Each Year: Percent of Total Exotic Cover



Playground and Deep Dive

Lost and gained native species over time

```
# Get native species from 2022
native_2022 <- alldata_22 %>%
  filter(cover_type == "Native") %>%
  distinct(species) %>%
  pull(species)

# Get native species from 2025
native_2026 <- alldata_now %>%
  filter(cover_type == "Native") %>%
  distinct(species) %>%
  pull(species)

# Find species in 2022 but not in 2025
species_lost <- setdiff(native_2022, native_2026)

species_lost

## [1] "Lupinus_succulentus"      "Eschscholzia_californica"
## [3] "Dudleya_lanceolata"      "Lupinus_microcarpus"
## [5] "Stebbinsoseris_heterocarpa" "Sidalcea_malviflora"
## [7] "Encelia_californica"     "Psilocarphus_brevissimus"
## [9] "Crassula_aquatica"       "Plagiobothrys_undulatus"
```

```
## [11] "Juncus_accutus"          "Eriogonum_parvifolium"
## [13] "Typha sp."              "Sparganium_uricarpum"
## [15] "Anemopsis_californica"

# Species in 2025 but not in 2022
species_gained <- setdiff(native_2026, native_2022)

species_gained

## [1] "Croton_setiger"          "Eleocharis_palustris"
## [3] "Spergularia_marina"      NA
## [5] "Grindelia_camporum"      "Sparganium_eurycarpum"
## [7] "Rosa_californica"        "Baccharis_salicifolia"
## [9] "Typha_sp."              "Juncus_accutus"
## [11] "Eleocharis_montivedensis" "Abronia_maritima"
```

Problem Transects Over Time

Up1, Wet 4, Bio: Native vs Exotic Composition

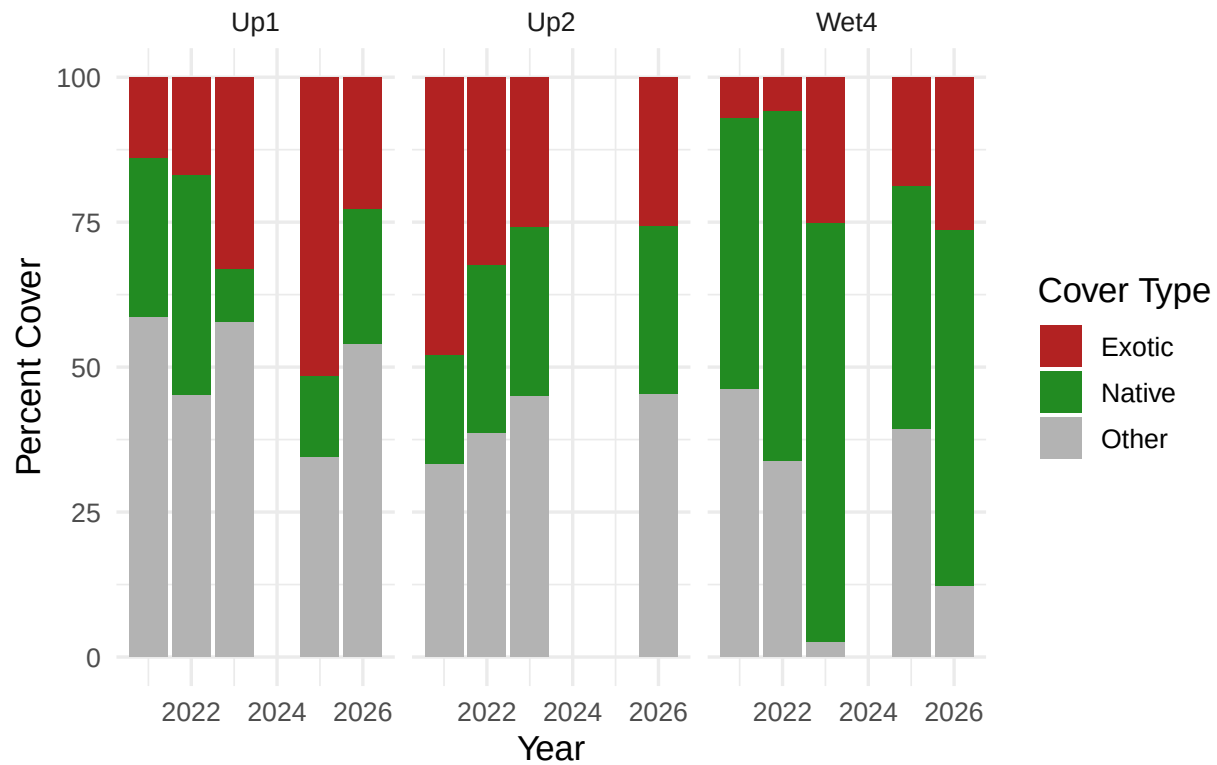
```
# Filter for the two transects of interest
focus_transects <- alldata_all %>%
  filter(transect %in% c("Up1", "Up2", "Wet4"))

# Summarize percent cover by year x transect x cover type
upbio_summary <- focus_transects %>%
  group_by(year, transect, cover_type) %>%
  summarize(total_cover = sum(cover, na.rm = TRUE), .groups = "drop") %>%
  group_by(year, transect) %>%
  mutate(percent_cover = (total_cover / sum(total_cover)) * 100)

# Only keep Native + Exotic for relative cover
upbio_relative <- upbio_summary %>%
  filter(cover_type %in% c("Native", "Exotic")) %>%
  group_by(year, transect) %>%
  mutate(percent_relative = (total_cover / sum(total_cover)) * 100)

### ---- Plot 1: Percent Cover Over Time (Up1, Wet4, Bio) ----
ggplot(upbio_summary, aes(x = year, y = percent_cover,
  fill = cover_type)) +
  geom_col(position = "stack") +
  facet_wrap(~ transect) +
  labs(
    title = "Total Percent Cover Over Time (Up1 & Bio)",
    y = "Percent Cover",
    x = "Year",
    fill = "Cover Type"
  ) +
  theme_minimal(base_size = 13) +
  scale_fill_manual(values = c(Native = "forestgreen",
    Exotic = "firebrick",
    Other = "gray70"))
```

Total Percent Cover Over Time (Up1 & Bio)



```
### ---- Plot 2: Relative Cover (Native vs Exotic Only) ----
ggplot(upbio_relative,
  aes(x = year, y = percent_relative, fill = cover_type)) +
  geom_col(position = "stack") +
  facet_wrap(~ transect) +
  labs(
    title = "Relative Vegetation Cover (Native vs Exotic Only)",
    y = "Percent of Vegetative Cover",
    x = "Year",
    fill = "Cover Type",
  ) +
  theme_minimal(base_size = 13) +
  theme(
    axis.text.x = element_text(size = 8) # or whatever size you want
  ) +
  scale_fill_manual(values = c(Native = "forestgreen",
    Exotic = "firebrick"))
```

